



Balancing Accuracy and Responsiveness in Real-Time Bird Sound Identification

Lea Frost

Computer Science, Data Science

Problem

Problem:

Real-time bird sound identification needs to provide fast and accurate predictions while operating under limited computing resources.

Audience:

Researchers, birders, and conservationists who use continuous real-time acoustic monitoring in noisy and resource-constrained environments.

ML Systems Perspective:

Continuous audio inference requires a tradeoff between latency, classification quality, model complexity, and resource usage.

Interest:

Interested in applying machine learning to real-time audio classification and exploring the required trade offs necessary to make it realistically usable. Also interested in birds!

Project type: Real World System/Tool

Research Question:

How do audio processing choices and model configuration affect the accuracy, latency, and computational efficiency of real-time bird sound identification?

Proposed Solution

Solution: Build and evaluate a full pipeline for real-time bird identification.



Primary Artifact:

Real-time bird identification system with benchmarking framework.

Existing Implementations:

PyTorch, torchaudio, librosa, and standard audio processing techniques/libraries.

Contribution:

Bird-sound classifier, real-time inference pipeline, benchmarking

Outside Project Scope:

Full production-ready deployment, mobile app, covering all species

Evaluation

Research Questions: How does model configuration and audio processing choices like window size and prediction frequency affect the accuracy, latency, and efficiency of a real-time bird identification system?

Independent Variables: Audio window size, prediction frequency, model configuration

Baseline: A standard PyTorch classifier with fixed audio configuration

Data: BirdCLEF + fixed subset of species

Rigor: Fixed data splits across experiments, identical preprocessing and hardware, multiple random seeds, repeated trials

Evidence: Evaluation metric table, model performance vs latency plot, resource usage comparison, potential real-time demo

Metrics

Quality

Accuracy
Precision
F1

System

Time-to-detection
Inference latency (P50/P95)
CPU/memory